Integration by Parts

Section 1: Standard Integration by Parts Formula

1 Evaluate $\int xe^{2x}dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$

b. $\frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{2}x^2e^{2x} + C$

2 Evaluate $\int \ln x dx$ for $x > 0$.

a. $x\ln x - x + C$

b. $x\ln x + x + C$

c. $\frac{1}{x} + C$

d. $\frac{(\ln x)^2}{2} + C$

3 Evaluate $\int_0^1 x\cos(\pi x)dx$.

a. $-\frac{2}{\pi^2}$

b. $\frac{2}{\pi^2}$

c. 0

d. $-\frac{1}{\pi}$

Section 2: Tabular Integration by Parts

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4 Using tabular integration, evaluate $\int x^2 e^x dx$.

a. $(x^2 - 2x + 2)e^x + C$

b. $(x^2 + 2x + 2)e^x + C$

c. $(x^2 - 2x)e^x + C$

d. $x^2 e^x - 2e^x + C$

5 Evaluate $\int x^3 \sin x dx$ using tabular integration.

a. $-x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \sin x + C$

b. $x^3 \cos x - 3x^2 \sin x + C$

c. $-x^3 \cos x - 3x^2 \sin x + C$

d. $-x^3 \sin x + 3x^2 \cos x + C$

6 When is tabular integration applicable?

a. When one factor is a polynomial that terminates to 0 upon repeated differentiation

b. Only for trigonometric functions

c. Only for logarithmic integrals

d. When both functions are exponential

Section 3: Cycling Integrals (Solving for the Unknown Integral)

7 Evaluate $\int e^x \cos x dx$.

a. $\frac{1}{2} e^x (\cos x + \sin x) + C$

b. $\frac{1}{2} e^x (\cos x - \sin x) + C$

c. $e^x \sin x + C$

d. $e^x (\cos x + \sin x) + C$

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8 Evaluate $\int e^{2x}\sin x dx$.

a. $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$

b. $\frac{e^{2x}}{5}(\sin x - 2\cos x) + C$

c. $\frac{e^{2x}}{4}(2\sin x - \cos x) + C$

d. $e^{2x}\cos x + C$

9 What is $\int \sec^3 x dx$?

a. $\frac{1}{2}(\sec x\tan x + \ln|\sec x + \tan x|) + C$

b. $\frac{1}{3}\sec^3 x + C$

c. $\sec x\tan^2 x + C$

d. $\frac{1}{2}\sec^2 x + C$

Section 4: Integration by Parts with Inverse Trigonometric Functions

10 Evaluate $\int \arctan x dx$.

a. $x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$

b. $x\arctan x + \frac{1}{2}\ln(1 + x^2) + C$

c. $\frac{1}{1 + x^2} + C$

d. $x\arctan x - \ln(1 + x) + C$

11 Evaluate $\int \arcsin x dx$.

a. $x\arcsin x + \sqrt{1 - x^2} + C$

b. $x\arcsin x - \sqrt{1 - x^2} + C$

c. $\frac{1}{\sqrt{1 - x^2}} + C$

d. $x\arcsin x + \frac{1}{2}\sqrt{1 - x^2} + C$

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12 Evaluate $\int_0^1 x\arctan x dx$.

a. $\frac{\pi}{4} - \frac{1}{2}$

b. $\frac{\pi}{8}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{8} - \frac{1}{4}$

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